



CS/CE 224/227: Object Oriented Programming and Design Methodologies

Week 02/Unit 02: Arrays

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Unit Outline

1. Review of Last Week's Content
2. The switch-case statement
3. Arrays in C++
4. Arrays and Functions
5. C-Strings



Introduction

- Last week, we covered the basics of C++.
- In particular, the following topics were covered: variables, data types, casting, conditional statements (if-then-else), loops (for-loop, while-loop, do-while-loop).
- We also introduced the increment operator ++?
- One of the questions posed in the class was:
- What is the difference between ++i and i++?



Pre-increment and Post-increment Operators

- `++i` is called the 'pre-increment' operator. `i++` is called the 'post-increment' operator.
- `++i` increments the value of its operand (`i`), before executing the statement.
- So, the output of the snippet:

```
int i=4; cout << "i=" << ++i; is  
i=5;
```

- What is the output of the snippet:

```
int i=4; cout << "i=" << i++;?
```

What is the output of the following program?

```
#include <iostream>
using namespace std;
int main(void) {
    int i = 4;
    cout << "i = " << (i++) << "\n";
    if (++i > 5 )
        cout << "i is equal to 6\n";
    cout << "i = " << i << "\n";
    return 0;
}
```



Recall the if-else-statement in C++

```
// Compare and find the maximum value
if (a >= b && a >= c) {
    max = a;
} else if (b >= a && b >= c) {
    max = b;
} else {
    max = c;
}
```



Let's walk through an Example

- Write a C++ program that reads an integer n and prints whether it is a prime number or not.
- Let's Code!
 - First Pseudo code on the board
 - Then we will run the program in VS Code

switch-case statement

- There is a special construct in C++ that uses multiple if statements.
- This is called the switch statement.
- The syntax of the switch-statement is:
 - The switch expression is evaluated once,
 - The value of the expression is compared with the values of each case,
 - If there is a match, the associated block of code is executed,
 - The break keyword is used to skip subsequent cases.
 - The default keyword is used to jump to the default case.

```
switch(expression) {  
    case x:  
        // code block  
        break;  
    case y:  
        // code block  
        break;  
    default:  
        // code block  
}
```

Switch and if-else side by side

- **If-else** checks condition sequentially

```
#include <iostream>
using namespace std;

int main(){

    int day = 3;

    if (day == 1) {
        cout << "Monday";
    }
    else if (day == 2) {
        cout << "Tuesday";
    }
    else if (day == 3) {
        cout << "Wednesday";
    }
    else {
        cout << "Invalid day";
    }

}
```

- **switch** directly jumps to the matching case (but only works with discrete values like int, char, enums).
- switch code is cleaner and faster when testing one variable against many constant values.

```
#include <iostream>
using namespace std;

int day = 3;

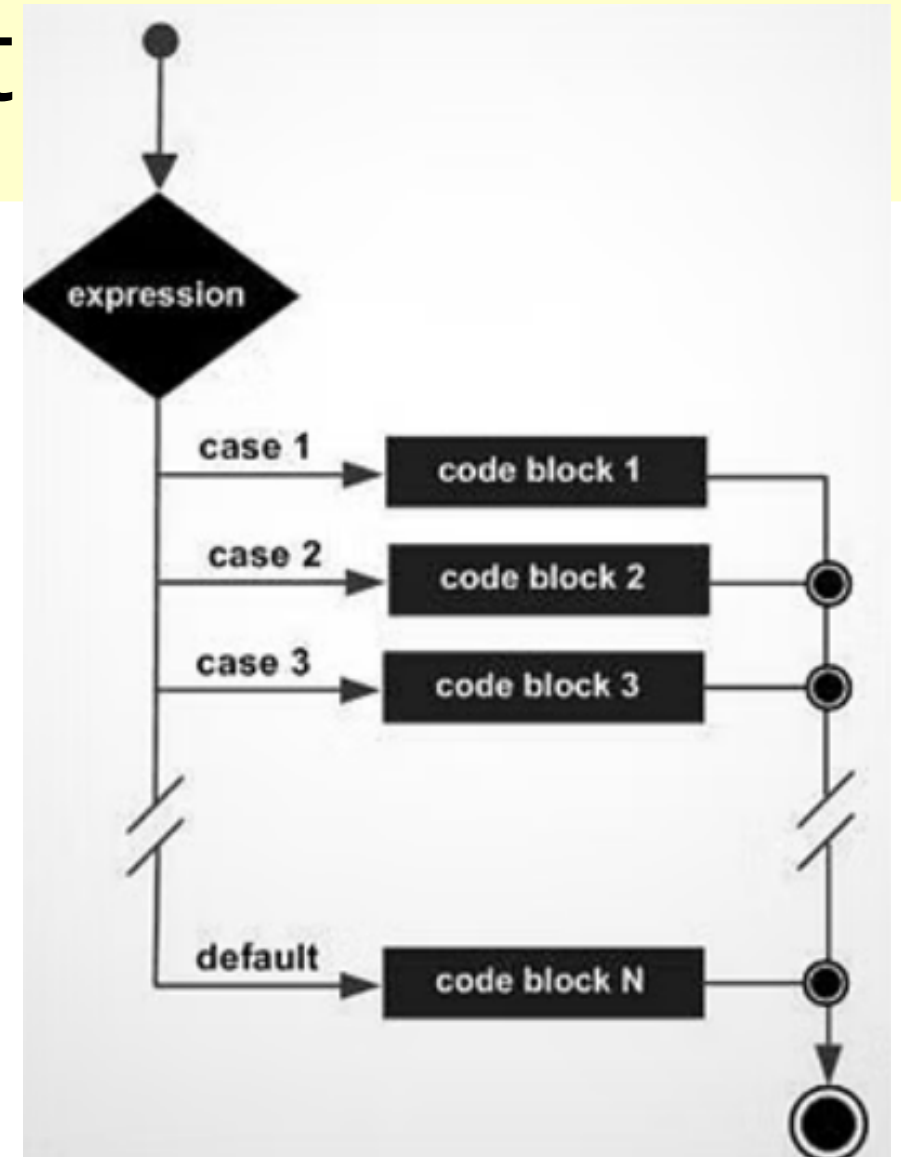
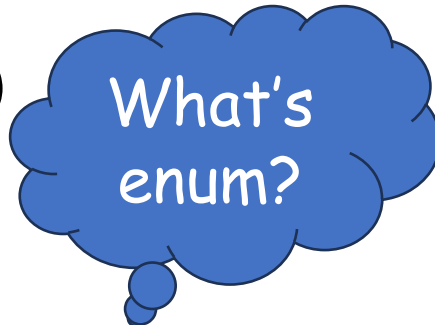
int main(){

    switch (day) {
        case 1:
            cout << "Monday";
            break;
        case 2:
            cout << "Tuesday";
            break;
        case 3:
            cout << "Wednesday";
            break;
        default:
            cout << "Invalid day";
    }

}
```


switch-case statement

- Flowchart for the switch-case statement shown on the right.
- The expression must evaluate to integer or constant values.
- Floating point or double precision values are not allowed.
- enumerated values (**enum**) may also be used for the expression.





switch - case Examples

- Design and implement a program in C++ that uses a switch statement to state whether an input integer is odd, even, prime or composite. Try using if-else for this program.
- The input must be between 2 and 9 (inclusive), else a default statement "out of range" is printed. Try using switch also.



switch-case Examples

- Design and implement a program in C++ that uses a switch statement to state whether an input integer is odd, even, prime or composite. Try using if-else for this program.
- The input must be between 2 and 9 (inclusive), else a default statement "out of range" is printed. Try using switch also.

```
int main() {
    int num;
    cout << "Enter an integer between 2 and 9: ";
    cin >> num;

    switch (num) {
        case 2:
            cout << num << " is even and prime." << endl;
            break;
        case 3:
        case 5:
        case 7:
            cout << num << " is odd and prime." << endl;
            break;
        case 4:
        case 6:
        case 8:
            cout << num << " is even and composite." << endl;
            break;
        case 9:
            cout << num << " is odd and composite." << endl;
            break;
        default:
            cout << "Out of range." << endl;
    }

    return 0;
}
```



Switch-Case Examples using enum

"Write a C++ program using **enum** and **switch** to display the number of days in a given month. The user inputs a number (1–12) representing the month, and the program outputs how many days that month has. Handle invalid inputs with a default case."



Switch-case Examples using enum

```
enum Months {  
    January = 1, February, March, April, May, June,  
    July, August, September, October, November, December  
};  
  
int main() {  
    int month;  
    cout << "Enter the month (1-12): ";  
    cin >> month;  
  
    switch (month) {  
        case January: case March: case May: case July:  
        case August: case October: case December:  
            cout << "31 days." << endl;  
            break;  
        case April: case June: case September: case November:  
            cout << "30 days." << endl;  
            break;  
        case February:  
            cout << "28 or 29 days (in a leap year)." << endl;  
            break;  
        default:  
            cout << "Invalid month. Please enter a number between 1 and 12." << endl;  
    }  
}
```



Switch-case Examples without using enum

```
#include <iostream>
using namespace std;

int main() {
    int month;
    cout << "Enter the month (1-12): ";
    cin >> month;

    switch (month) {
        case 1: case 3: case 5: case 7:
        case 8: case 10: case 12:
            cout << "31 days." << endl;
            break;

        case 4: case 6: case 9: case 11:
            cout << "30 days." << endl;
            break;

        case 2:
            cout << "28 or 29 days (in a leap year)." << endl;
            break;

        default:
            cout << "Invalid month. Please enter a number between 1 and 12." << endl;
    }
    return 0;
}
```



Arrays

- An array is a data type containing elements of the same type (homogeneous) data, stored in contiguous memory locations.
- We declare an array by specifying the data type, followed by the name of the array, followed by its size in (square brackets).
- `int list[4]` declares an array of 4 integers called `list`.

list[0]	list[1]	list[2]	list[3]
----------------	----------------	----------------	----------------

- Each element of the array can be accessed by its subscripts.
- Subscripts start from 0 to size – 1 (in this case from 0 to 3).

```
//Some array
//initialization options

int a[4];
//declares an
//array of size 4

int a[] = {20, 40};
//declares an array
//with values
//size is implicit

int a[];
//illegal; size must
//be specified
```

Arrays in C++

- Arrays in C++ can be one-dimensional or multi-dimensional.
- A declaration of a two dimensional array is given as follows:
- `dataType arrayName[rows][columns];`
- `double m[3][4];` *//this declares an array of size 12 i.e. 3 rows and 4 columns.*
- The array elements are indexed as follows:

m[0] [0]	m[0] [1]	m[0] [2]	m[0] [3]
m[1] [0]	m[1] [1]	m[1] [2]	m[1] [3]
m[2] [0]	m[2] [1]	m[2] [2]	m[2] [3]

```
//LEGAL WAYS TO  
//DECLARE A 2-D ARRAY  
int m[][4] = {{1, 2, 3, 4},  
              {5, 6, 7, 8},  
              {9, 10, 11, 12}};  
  
int m[3][4];
```


Processing arrays in C++

- We use loops (e.g. for-loops) to process arrays in C++. To print a 1-D array:

```
int a[] = {1, 2, 3, 4};  
for(int i = 0; i < 4; i++)  
    cout << a[i] << " ";
```

- To process a 2-D array, we use nested for-loops. To print a 2-D array:

```
int a[][2] = {{1, 2}, {3, 4}};  
for(int i = 0; i < 2; i++) {  
    for(int j = 0; j < 2; j++)  
        cout << a[i][j] << " ";  
    cout << "\n";  
}
```



Processing arrays in C++

- Question: What is a 2D array?
- Question: What is an ND-Array?



Processing arrays in C++

- Question: What is a 2D array?
 - **Matrix**
- Question: What is an ND-Array?
 - **Tensor** (A trivia, not important for this class)
 - See pytorch



Arrays and Functions (Without pointers)

- Arrays can be passed as parameters to a function.
- However, the function has to be declared first.
- This is necessary before the first use of the function in the main function.
- A function declaration can be done by declaring the function by stating its return type, followed by name, followed by the signatures (parameters and their types).
- `int power(int base, int exponent)` is a function declaration.
- In C++ a function definition (the header and entire body of the function) can also serve as a declaration, if it is stated before the main function.
- Examples of function declaration with arrays:



Arrays and Functions

```
#include <iostream>
using namespace std;

// Constants for array dimensions
const int DISTRICTS = 4;
const int MONTHS = 3;

// Function declaration (prototype)
void display(double sales[DISTRICTS][MONTHS]);

int main() {
    // Initialize 2D array (sales data)
    double sales[DISTRICTS][MONTHS] = {
        {1200.5, 1300.0, 1250.75},
        {1500.25, 1600.0, 1550.5},
        {1700.0, 1750.25, 1800.75},
        {1900.0, 1950.5, 2000.0}
    };

    // Call function to display array
    display(sales);

    return 0;
}
```

- Observe the use of const to declare constants DISTRICTS and MONTHS
- Question: What will this code do?
- Do we need something else?

Arrays and Functions

```
// Function definition
void display(double sales[DISTRICTS][MONTHS]) {
    cout << "Sales Report (Districts x Months)\n";

    for (int i = 0; i < DISTRICTS; i++) {
        cout << "District " << (i + 1) << ": ";
        for (int j = 0; j < MONTHS; j++) {
            cout << sales[i][j] << " ";
        }
        cout << endl;
    }
}
```

- What is that something else?
- Actual function definition
- Declaration without definition **does nothing**



Arrays and Functions

Function Declaration with Array Arguments

- In function declarations, array parameters are written with **data type and size**.
- Example (our sales data case):
- **void display(double sales[DISTRICTS][MONTHS]); // declaration**

The **row size is not required** in the declaration (compiler can deduce it).

- **void display(double sales[][MONTHS]); // declaration**

When passing **1D arrays**, the size is also not required in the declaration.

- **void printArray(int arr[]); // declaration**



Arrays and Functions – function calls

Function Call with Array Arguments

When the function is called, only the name of the array is used as an argument.

```
display(sales);    // function call
```

Example: 1-D Array Addition and Multiplication.
Write a program in C++ that reads in as inputs two arrays of one dimension using a function read. It then computes the element wise sum and product of these two arrays using the functions sum and product. Make sure to use suitable parameter lists for the functions.



Arrays and Functions – function calls

```
#include <iostream>
using namespace std;

// Function prototypes
void read(int arr[], int size);
void sum(const int arr1[], const int arr2[], int result[], int size);
void product(const int arr1[], const int arr2[], int result[], int size);
void display(const int arr[], int size);

// Reads array elements
void read(int arr[], int size) {
    for (int i = 0; i < size; i++) {
        cout << "Element " << i+1 << ": ";
        cin >> arr[i];
    }
}

// Displays array elements
void display(const int arr[], int size) {
    for (int i = 0; i < size; i++) {
        cout << arr[i] << " ";
    }
    cout << endl;
}
```

```
int main() {
    const int SIZE = 5; // Example size
    int arr1[SIZE], arr2[SIZE], sumArr[SIZE], prodArr[SIZE];

    cout << "Enter elements for first array:\n";
    read(arr1, SIZE);

    cout << "Enter elements for second array:\n";
    read(arr2, SIZE);

    // Display results
    cout << "\nFirst Array: ";
    display(arr1, SIZE);
}
```

Arrays and Functions – function calls

```
#include <iostream>
using namespace std;

// Function prototypes
void read(int arr[], int size);
void sum(const int arr1[], const int arr2[], int result[], int size);
void product(const int arr1[], const int arr2[], int result[], int size);
void display(const int arr[], int size);

// Reads array elements
void read(int arr[], int size) {
    for (int i = 0; i < size; i++) {
        cout << "Element " << i+1 << ": ";
        cin >> arr[i];
    }
}

// Displays array elements
void display(const int arr[], int size) {
    for (int i = 0; i < size; i++) {
        cout << arr[i] << " ";
    }
    cout << endl;
}
```

Declaration

Definition

```
int main() {
    const int SIZE = 5; // Example size
    int arr1[SIZE], arr2[SIZE], sumArr[SIZE], prodArr[SIZE];

    cout << "Enter elements for first array:\n";
    read(arr1, SIZE);

    cout << "Enter elements for second array:\n";
    read(arr2, SIZE);

    // Display results
    cout << "\nFirst Array: ";
    display(arr1, SIZE);
}
```

Execution



Arrays and Functions

- In an a function declaration, all (but one) of the subscripts of the array size must be stated.
- So for example,
- `void f(int [][] a)` is not legal, but `void f(int a[][COLS])` is legal.
- Likewise for a 3 dimensional array, `void f(int a[][WIDTH][DEPTH])` is legal.
- It is advisable to pass the length of the array as a parameter, or declare it as a global constant using `#define` primitive.



1-D Arrays as Strings

- Two kinds of strings used in C++:
 - Objects of the `string` class.
 - One dimensional arrays of type `char`, also known as C-strings.
- A C-string can be defined as an array of characters.
- The length of the string has to be declared.
- Usually it is the maximum possible size the string is supposed to have.
- Here's one way to declare a C-string:

```
const int MAX = 80;  
char str[MAX];
```



C-Strings: Example Program

```
#include <iostream>

using namespace std;

int main(void) {
    const int MAX = 80;
    char str[MAX];

    cout << "Enter the string: ";
    cin >> str;

    cout << "The string you entered: " << str << endl;
    return 0;
}
```



C-Strings

C	O	M	P	U	T	E	R	\0	
---	---	---	---	---	---	---	---	----	--

- Each character occupies 1 byte of memory.
- C-strings must terminate with a byte containing 0.
- This is represented by the character constant `'\0'`, which has an ASCII value of 0. This terminating zero is called the null character.
- The `<<` operator displays characters until it encounters the null character.



C-Strings: Some pitfalls to avoid

- To avoid buffer overflow (what is a buffer?) while reading strings, use the `setw` operator. (Page 292, Lafore)
- You can initialize strings by directly assigning the string to the array (Page 292):

```
char str[] = "Hello! How are you?"
```

- To read text containing blanks using `cin`, we use another function, `cin.get()`.
- To read a string with multiple lines, use `cin::get()` with a terminating character.
- To copy a string character by character, use a for-loop.



C-Strings: Library Functions

- Some important C-String functions are given below.
- The #include file is cstring:

<u>strcat()</u>	Appends one C-style string to the end of another
<u>strcmp()</u>	Compares the ASCII values of characters in two C-style strings to determine which string has a higher value
<u>strcpy()</u>	Copies the characters of a C-style string into the memory of another string
<u>strlen()</u>	Return the length of a C-style string



C-Strings: Library Functions

- Some important C-String functions are given below.
- The #include file is cstring:

strcat()

```
char str1[20] = "Hello ";
char str2[] = "World!";
strcat(str1, str2);    // appends str2 to str1
cout << str1 << endl;  // Output: Hello World!
```

strcpy()

```
char source[] = "OpenAI";
char destination[20];

strcpy(destination, source);
cout << "Copied string: " << destination << endl;
// Output: Copied string: OpenAI
```

strcmp()

```
char str1[] = "Apple";
char str2[] = "Banana";

if (strcmp(str1, str2) == 0)
    cout << "Strings are equal" << endl;
else if (strcmp(str1, str2) < 0)
    cout << str1 << " comes before " << str2 << endl;
else
    cout << str1 << " comes after " << str2 << endl;
// Output: Apple comes before Banana
```



Arrays of Strings

```
#include <iostream>
using namespace std;

int main()
{
    const int DAYS = 7;           //number of strings in array
    const int MAX = 10;           //maximum size of each string
                                   //array of strings
    char star[DAYS][MAX] = { "Sunday", "Monday", "Tuesday",
                              "Wednesday", "Thursday",
                              "Friday", "Saturday" };
    for(int j=0; j<DAYS; j++)      //display every string
        cout << star[j] << endl;
    return 0;
}
```



Summary

- In this slideset we studied switch statements with options for default, break.
- We also studied arrays and parameter passing using arrays and functions.
- Finally we studied the use of 1-D arrays as strings.
- It needs to be mentioned here that C++ does not have methods for checking array bounds, i.e. accessing an element beyond the array boundaries is not checked.
- For this reason, due care should be exercised while accessing array elements.